



ENGINEERING MATHEMATICS-II

Department of Science and Humanities

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UNIT 1: INTEGRAL CALCULUS

Class 10: change of variables in
triple integral, SPHERICAL
coordinates

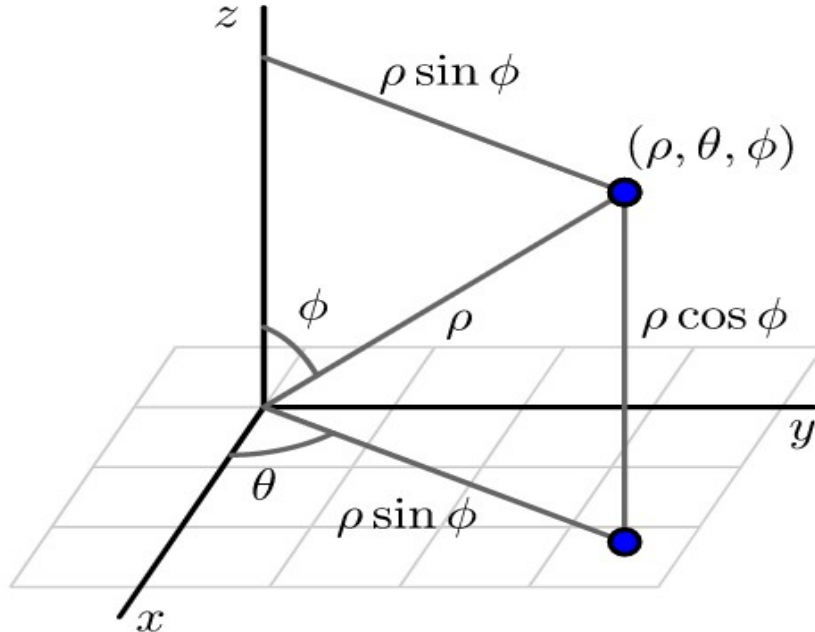
TRIPLE INTEGRAL IN SPHERICAL COORDINATES: MOTIVATING QUESTIONS



- What are the spherical coordinates of a point, and how are they related to Cartesian coordinates?
- What is the volume element in spherical coordinates? How does this inform us about evaluating a triple integral as an iterated integral in spherical coordinates?

TRIPLE INTEGRAL IN SPHERICAL COORDINATES:

the spherical coordinates of a point in 3- space have the form (ρ, θ, ϕ) , where ρ is the distance from the point to the origin, θ has the same meaning as in polar coordinates, and ϕ is the angle between the positive z axis and the vector from the origin to the point.



TRIPLE INTEGRAL IN SPHERICAL COORDINATES:

- The spherical coordinates of a point P in 3-space are ρ (rho), θ , and ϕ (phi), where ρ is the distance from P to the origin, θ is the angle that the projection of P onto the xy -plane makes with the positive x -axis, and ϕ is the angle between the positive z axis and the vector from the origin to P . When P has Cartesian coordinates (x, y, z) , the spherical coordinates are given by

$$\rho^2 = x^2 + y^2 + z^2, \quad \tan(\theta) = \frac{y}{x}, \quad \cos(\phi) = \frac{z}{\rho}.$$

Given the point P in spherical coordinates (ρ, ϕ, θ) , its rectangular coordinates are

$$x = \rho \sin(\phi) \cos(\theta), \quad y = \rho \sin(\phi) \sin(\theta), \quad z = \rho \cos(\phi).$$

- The volume element dV in spherical coordinates is $dV = \rho^2 \sin(\phi) d\rho d\theta d\phi$. Thus, a triple integral $\iiint_S f(x, y, z) dA$ can be evaluated as the iterated integral

$$\iiint_S f(\rho \sin(\phi) \cos(\theta), \rho \sin(\phi) \sin(\theta), \rho \cos(\phi)) \rho^2 \sin(\phi) d\rho d\theta d\phi.$$

TRIPLE INTEGRAL IN SPHERICAL COORDINATES:

For spherical coordinates, the change of variables function is

$$(x, y, z) = \mathbf{T}(\rho, \theta, \phi)$$

where the components of $\mathbf{T}$ are given by

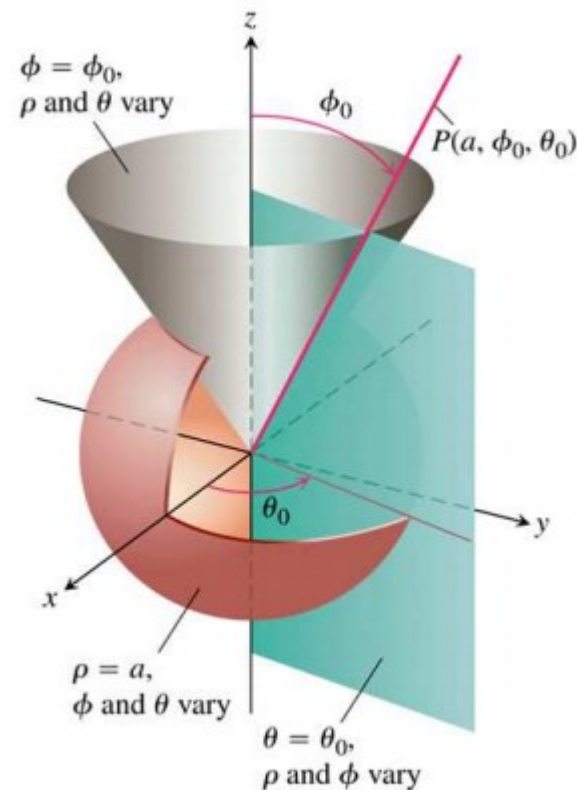
$$\begin{aligned}x &= \rho \sin \phi \cos \theta \\y &= \rho \sin \phi \sin \theta \\z &= \rho \cos \phi.\end{aligned}$$

We can compute that

$$\begin{aligned}D\mathbf{T}(\rho, \theta, \phi) &= \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \phi} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \phi} \end{vmatrix} \\&= \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta & \rho \cos \phi \cos \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta & \rho \cos \phi \sin \theta \\ \cos \phi & 0 & -\rho \sin \phi \end{vmatrix} \\&= -\rho^2 \sin \phi.\end{aligned}$$

The change of variable factor is the absolute value of the determinant

$$|D\mathbf{T}(\rho, \theta, \phi)| = \rho^2 \sin \phi.$$



VOLUME USING TRIPLE INTEGRAL IN SPHERICAL COORDINATES:

The Volume Element in Spherical Coordinates

$$dV = \rho^2 \sin \varphi \, d\rho \, d\theta \, d\varphi$$

Finding the Volume of l'Hemisphèric

Hot air balloons



EXAMPLE 1: VOLUME USING TRIPLE INTEGRAL IN SPHERICAL COORDINATES

Finding the Volume of an Ellipsoid

Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

We again use symmetry and evaluate the volume of the ellipsoid using spherical coordinates.

As before, we use the first octant $x \geq 0, y \geq 0$, and $z \geq 0$ and then multiply the result by 8.

In this case the ranges of the variables are

$$0 \leq \varphi \leq \frac{\pi}{2}, 0 \leq \rho \leq \frac{\pi}{2}, 0 \leq \rho \leq 1, \text{ and } 0 \leq \theta \leq \frac{\pi}{2}.$$

EXAMPLE 1: VOLUME USING TRIPLE INTEGRAL IN SPHERICAL COORDINATES

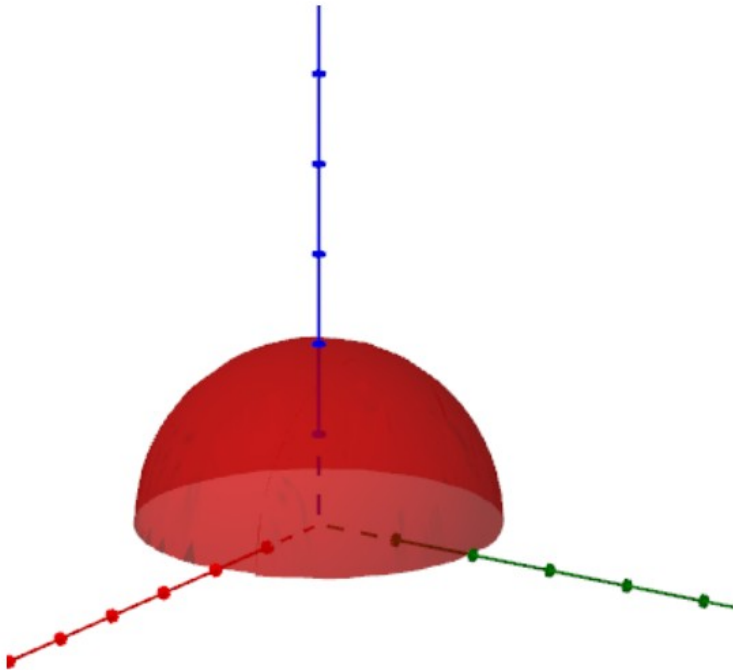
Then the volume of the ellipsoid becomes

$$\begin{aligned} V &= \iiint_D dx \, dy \, dz \\ &= 8 \int_{\theta=0}^{\theta=\pi/2} \int_{\rho=0}^{\rho=1} \int_{\varphi=0}^{\varphi=\pi/2} abc \rho^2 \sin \theta \, d\varphi \, d\rho \, d\theta \\ &= 8abc \int_{\varphi=0}^{\varphi=\pi/2} d\varphi \int_{\rho=0}^{\rho=1} \rho^2 d\rho \int_{\theta=0}^{\theta=\pi/2} \sin \theta \, d\theta \\ &= 8abc \left(\frac{\pi}{2} \right) \left(\frac{1}{3} \right) (1) \\ &= \frac{4}{3} \pi abc. \end{aligned}$$

EXAMPLE 2: EVALUATION TRIPLE INTEGRAL IN SPHERICAL COORDINATES

Evaluate $\int \int \int_E e^{(x^2+y^2+z^2)^{3/2}} dV$ where E is below $x^2 + y^2 + z^2 \leq 1$ (often referred to as top half of the unit ball) and above $z = 0$.

1. A sketch will be helpful here.



$$E = \begin{cases} 0 \leq \rho \leq 1 \\ 0 \leq \theta \leq 2\pi \\ 0 \leq \phi \leq \pi/2 \end{cases}$$

EXAMPLE 2: EVALUATION TRIPLE INTEGRAL IN SPHERICAL COORDINATES

2. Rewrite the function into spherical coordinates

$$e^{(x^2+y^2+z^2)^{3/2}} = e^{(\rho^2)^{3/2}} = e^{\rho^3}$$

3. Set up the integral

$$\int_0^1 \int_0^{2\pi} \int_0^{\pi/2} e^{\rho^3} \cdot \rho^2 \sin(\phi) \, d\phi \, d\theta \, d\rho$$

4. Note that the integrand is a product of functions of ϕ , ρ , and θ . This means we can rewrite the integral as

$$\int_0^1 \rho^2 e^{\rho^3} \, d\rho \cdot \int_0^{2\pi} 1 \, d\theta \cdot \int_0^{\pi/2} \sin(\phi) \, d\phi$$

EXAMPLE 2: EVALUATION TRIPLE INTEGRAL IN SPHERICAL COORDINATES

$$(a) \int_0^1 \rho^2 e^{\rho^3} d\rho$$

i. Let $u = \rho^3$

$$\text{ii. } du = 3\rho^2 d\rho \Rightarrow \frac{1}{3} du = \rho^2 d\rho$$

$$\text{iii. If } \rho = 0, u = (0)^3 = 0$$

$$\text{iv. If } \rho = 1, u = (1)^3 = 1$$

v. Substitute

$$\int_0^1 \frac{1}{3} e^u du = \frac{1}{3} e^u \Big|_0^1 = \frac{1}{3} e - \frac{1}{3}$$

EXAMPLE 2: EVALUATION TRIPLE INTEGRAL IN SPHERICAL COORDINATES

$$(b) \int_0^{2\pi} 1 \, d\theta = \int_0^{2\pi} 1 \, d\theta = \theta \Big|_0^{2\pi} = 2\pi$$

$$(c) \int_0^{\pi/2} \sin(\phi) \, d\phi = \int_0^{2\pi} 1 \, d\theta = \theta \Big|_0^{2\pi} = 2\pi$$

(d) Finally, multiply the three integral values together

$$\int_0^1 \int_0^{2\pi} \int_0^{\pi/2} e^{\rho^3} \cdot \rho^2 \sin(\phi) \, d\phi \, d\theta \, d\rho = \left(\frac{1}{3}e - \frac{1}{3} \right) \cdot 2\pi \cdot 1 = \frac{2}{3}\pi e - \frac{2}{3}\pi$$

EXAMPLE 3: AVERAGE TEMPERATURE USING TRIPLE INTEGRAL IN SPHERICAL COORDINATES

Suppose the temperature at (x, y, z) is $T = 1/(1 + x^2 + y^2 + z^2)$.

Find the average temperature in the unit sphere centered at the origin.

In two dimensions we add up the temperature at "each" point and divide by the area; here we add up the temperatures and divide by the volume, $(4/3)\pi$:

$$\frac{3}{4\pi} \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} \frac{1}{1 + x^2 + y^2 + z^2} dz dy dx$$

EXAMPLE 3: AVERAGE TEMPERATURE USING TRIPLE INTEGRAL IN SPHERICAL COORDINATES

This looks quite messy; since everything in the problem is closely related to a sphere, we'll convert to spherical coordinates.

$$\frac{3}{4\pi} \int_0^{2\pi} \int_0^\pi \int_0^1 \frac{1}{1+\rho^2} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{3}{4\pi} (4\pi - \pi^2) = 3 - \frac{3\pi}{4}.$$



THANK YOU
